

Chapter 24: From Application to Belonging: A Human-in-the-Loop Framework for AI-Assisted Advising in Indian Student Mobility

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Abstract

Generative artificial intelligence has been widely proposed as a remedy for the chronic under-capacity of academic advising in higher education. The existing literature, however, concentrates almost exclusively on advising that occurs after enrolment and within a single institution's boundaries: course selection, degree planning, and campus service navigation. It largely overlooks a longer and considerably higher-stakes advisory journey undertaken by internationally mobile students, one that begins with aspiration in the home country and ends, if it succeeds, in cultural belonging and employment abroad. This chapter addresses that gap. Drawing on reflective practice in cross-border educational consulting for Indian students entering the United States, it proposes a five-stage Mobility Advising Lifecycle and, within it, the AADD triage framework (Automate, Augment, Defer, Deny), which classifies advisory queries along two dimensions: regulatory volatility and consequence severity. The framework's central claim is that the appropriate role for AI in advising is not a function of question difficulty but of what happens when the answer is wrong. The chapter situates this argument at the intersection of educational technology, migration studies, and governance, and offers design guidance for institutions, consultancies, and policymakers.

Keywords: Artificial intelligence in education; Academic advising; International student mobility; Generative AI; Human-in-the-loop systems; AI governance; Higher education

1. Introduction

For a student in a Tier-2 Indian city preparing to study abroad, the decisions that matter most are rarely academic. Choosing between two universities is comparatively simple. Interpreting a financial affidavit, understanding what may lawfully be said in a visa interview, estimating the true cost of a first month in an unfamiliar city, or recognising the early signs of isolation in a shared apartment eight thousand miles from home: these are the problems that determine outcomes. They are also precisely the problems that formal advising infrastructure handles least well.

The arrival of generative AI has prompted considerable optimism that this advisory deficit can be closed at scale. That optimism is not unfounded. But it rests on a body of research whose object of study is narrower than the enthusiasm suggests. Studies of AI advising overwhelmingly examine institutional chatbots serving enrolled students within one university's informational domain, where the ground truth is stable, the institution controls the corpus, and the cost of an incorrect answer is a scheduling inconvenience. Cross-border advising inverts every one of these conditions. The relevant facts are set by immigration authorities rather than institutions, they change without notice, and an incorrect answer can terminate a student's legal status.

This chapter argues that the useful question is not whether AI can support student mobility advising, but which classes of advisory query it should be permitted to answer, and under what supervision. It develops an answer in three moves. Section 5 maps the advisory journey into five stages with distinct informational characteristics. Section 6 introduces the AADD triage framework, which assigns each query class an automation posture derived from regulatory volatility and consequence severity. Sections

7 and 8 examine the risks the framework is designed to contain and its implications for practice and policy.

2. Context: Indian Student Mobility Under Pressure

India is among the world's largest sources of internationally mobile students. Data placed before Parliament by the Ministry of External Affairs recorded approximately 1.25 million Indian students enrolled in higher education abroad, part of a wider total of some 1.88 million once school-level enrolment is counted, spread across more than 150 countries (Ministry of External Affairs, 2025). Growth in the higher education segment had been steep and sustained, rising from roughly 750,000 in 2022 to a peak near 1.33 million in 2024, before contracting by 5.7 per cent in 2025, the first decline after three consecutive years of expansion (ICEF Monitor, 2025).

That reversal matters for the argument developed here, because of what caused it. The decline was driven less by diminished aspiration than by an abrupt tightening of the regulatory and economic environment across destination countries: restrictive visa policy, elevated refusal rates, rising tuition, and currency depreciation (The PIE News, 2025). In other words, the informational terrain on which advising operates shifted faster than any advisory system could be updated.

This has a direct methodological consequence. A model trained on a corpus reflecting 2023 immigration conditions and asked in 2026 about eligibility, processing timelines, or post-study work rights will answer fluently and may answer wrongly, not because it reasoned poorly, but because the world moved. Volatility, not complexity, is the binding constraint. Any framework for AI-assisted mobility advising must therefore treat the rate of change of the underlying facts as a first-class design variable.

3. Related Work

3.1 AI in academic advising

Systematic reviews of chatbots in education document rapid growth in deployment and a consistent set of claimed benefits: continuous availability, reduction of routine advisor workload, and scalable personalisation (Okonkwo & Ade-Ibijola, 2021; Labadze et al., 2023). Akiba and Fraboni (2023) assess large language models specifically as advising instruments and reach a qualified conclusion that recurs throughout the literature: such systems handle general and procedural inquiries competently while remaining unreliable in complex, contingent advising scenarios. Domain-specific implementations, including bilingual advising agents built for multilingual student populations, report comparable patterns of strength and limitation (Bilquise et al., 2022).

3.2 Ethical and epistemic concerns

A parallel literature examines the risks. Kooli (2023) surveys ethical implications including opacity, accountability gaps, and the erosion of professional judgement. Concerns regarding bias and the confident dissemination of inaccurate information are widely reported (Ilieva et al., 2023). Meta-analytic evidence on learning outcomes remains mixed rather than uniformly positive (Wu & Yu, 2024), which counsels against treating deployment as self-evidently beneficial.

3.3 The gap

Two bodies of scholarship remain largely unconnected. Research on international student wellbeing has long documented the acculturative stress, isolation, and help-seeking reluctance that characterise the transition abroad (Mori, 2000), and the sociology of education explains differential outcomes partly through unequal distributions of cultural capital, the tacit knowledge of how systems actually work (Bourdieu, 1986). Tinto's (1993) account of persistence identifies integration and belonging, not information supply, as the determinants of whether students stay. Yet the AI advising literature is silent on the pre-departure and settlement phases where these forces are strongest. The framework proposed below is an attempt to bridge that divide.

4. Positioning and Approach

This chapter is a reflective practitioner contribution rather than an empirical study. Its observations derive from sustained cross-border advising practice with Indian students transitioning to the United States, conducted over more than a decade and comprising several thousand advisory relationships and over a thousand individual consultations. That experience is treated here as a source of structured hypotheses about where advisory failure concentrates, not as data.

The limits of this stance should be stated plainly. No controlled comparison is offered, no instrument was administered, and no claim to statistical generalisability is made. Practitioner recall is selective, and cases that ended badly are more memorable than those that did not. What such positioning can legitimately contribute is a taxonomy grounded in observed failure modes, a specification of the problem that subsequent empirical work can test. The framework in Section 6 is offered in that spirit, and Section 9 sets out how it might be validated.

5. The Mobility Advising Lifecycle

The advisory journey divides into five stages. They differ not merely in subject matter but in the stability of the relevant facts, the severity of error, and the emotional register in which questions are asked. These three properties govern how automation should be applied.

Table 1

The five stages of the mobility advising lifecycle

Stage	Representative queries	Dominant risk	Fact stability
1. Aspiration and selection	Field and destination choice; institutional reputation; return on investment	Poorly matched enrolment	Moderate
2. Application and documentation	Statements of purpose; transcript equivalence; recommendation logistics	Rejection; misrepresentation	Moderate to high

Stage	Representative queries	Dominant risk	Fact stability
3. Visa and financial compliance	Eligibility; proof of funds; interview conduct; status maintenance	Refusal; loss of lawful status	Low (volatile)
4. Arrival and settlement	Housing; banking; healthcare; documentation on landing	Financial harm; exploitation	Moderate
5. Belonging and career transition	Community; isolation; work authorisation; professional networks	Attrition; mental health decline	Mixed

Two features of this distribution are consequential. First, risk is not evenly spread: Stage 3 concentrates both the highest error severity and the lowest fact stability, an intersection where autonomous generation is least defensible. Second, Stage 5, where attrition and wellbeing outcomes are actually determined, is the stage least amenable to informational solutions of any kind, automated or otherwise. Belonging is produced by community, not by correct answers. Advising systems that measure themselves on query resolution will systematically under-serve the stage that matters most.

6. The AADD Triage Framework

Existing guidance on AI in advising tends to sort questions by difficulty, reserving human attention for the hard ones. In the mobility context this heuristic misleads. The question of whether a particular visa category permits a specific form of employment is not intellectually hard; it is simply volatile and consequential. Difficulty is the wrong axis.

AADD therefore classifies advisory queries along two other dimensions. Regulatory volatility is the rate at which the correct answer changes independently of the student's circumstances. Consequence severity is the magnitude of harm if the answer is wrong and acted upon. Crossing the two yields four automation postures.

Table 2

The AADD triage framework for AI-assisted mobility advising

Posture	Condition	Description	Example query
Automate	Low volatility, low severity	Direct AI response is appropriate; monitoring for accuracy suffices.	How does the US credit-hour system compare with Indian semester credits?

Posture	Condition	Description	Example query
Augment	Low volatility, high severity	AI drafts or structures; a qualified human reviews before the student acts.	Please review the framing and structure of my statement of purpose.
Defer	High volatility, low severity	System declines to state the fact and routes the student to the authoritative live source.	What is the current application fee and deadline for this intake?
Deny	High volatility, high severity	No generated answer under any framing; escalation to a licensed or accountable human is mandatory.	Given my status, may I take this off-campus internship?

The Deny quadrant is the framework's substantive commitment and its principal departure from prevailing practice. Contemporary systems handle this category badly in a specific way: rather than refusing, they hedge. A response that summarises the general rule and appends a recommendation to consult an advisor appears responsible while functioning as an answer, because students act on the summary and disregard the caveat. Partial refusal is not refusal. In the Deny quadrant the correct system behaviour is to decline substantive content entirely and to make escalation the only available path.

Deferral is the framework's second load-bearing element and the more easily implemented. Where a fact is volatile but the stakes are modest, the system should not attempt to know it. Retrieval from a governed, current source (or a direct handoff to the institution or consular authority that owns the fact) dominates generation from parametric memory. This is an architectural instruction as much as a policy one: mobility advising systems should be built to route rather than to recall.

AADD also clarifies what automation is genuinely for. The Automate and Augment quadrants together cover a substantial share of advisory volume, and displacing that volume is valuable precisely because it liberates scarce human attention for Stages 3 and 5, where judgement, accountability, and relationship are irreplaceable. The framework is not a brake on automation. It is an argument about where automation should be aimed.

7. Risks and Ethical Considerations

7.1 Fluent inaccuracy in compliance contexts

The characteristic failure of a language model in this domain is not incoherence but confident obsolescence. Advice reflecting a superseded regulatory regime is delivered in the register of authority, and the student has no basis for detecting the error. Because consequences may not surface for months, the feedback loops that would ordinarily correct such a system are absent.

7.2 Asymmetry of verification capacity

Students best positioned to identify a flawed answer, namely those with family precedent abroad, English fluency, and access to professional networks, are those least dependent on automated advising. Students from families without migration experience are simultaneously the most reliant and the least equipped to verify. Deployed without triage, AI advising risks widening the very gap in cultural capital it is promoted as narrowing (Bourdieu, 1986).

7.3 Privacy and data governance

Mobility advising conversations contain passport details, financial records, health information, and immigration history, all categories subject to divergent and sometimes conflicting protection regimes in the origin and destination jurisdictions. Cross-border processing of such data by third-party model providers raises consent and retention questions that most consultancies and many institutions have not addressed.

7.4 Substitution for human relationship

The final risk is the most consequential and the least measurable. Isolation and reluctance to seek help are established features of the international student transition (Mori, 2000), and disclosure of distress typically occurs indirectly, embedded within an ostensibly practical question. An always-available conversational system may absorb such disclosures without recognising them, and in doing so displace the human contact through which they would have been noticed. Efficiency in query handling can coexist with degradation in care.

8. Implications for Practice and Policy

For institutions and consultancies, AADD is directly operational. Advisory query logs can be classified into the four quadrants, and the Deny set implemented as hard architectural refusal rather than as prompt-level instruction, since instruction-based guardrails are known to be circumventable. Deferral demands investment in retrieval from governed sources rather than in model capability.

For destination-country institutions, the framework implies that machine-readable, authoritative, versioned publication of compliance information is a student-safety measure. Where such sources do not exist, generation fills the vacuum.

For policymakers, mobility advising merits recognition as a distinct AI application domain with its own risk profile, in which the harm accrues to a population that is non-resident, non-citizen, and structurally without recourse. Emerging governance frameworks that classify AI systems by risk should account for advisory systems whose errors bear on immigration status.

9. Limitations and Future Research

The framework is conceptual and awaits validation. Three studies would test it. First, an audit study submitting matched query sets drawn from each quadrant to widely used general-purpose models, scoring responses for accuracy against contemporaneous regulation and for appropriateness of refusal. Second, a comparative trial of AADD-governed and ungoverned advising channels, measuring both resolution efficiency and downstream compliance incidents. Third, qualitative work on how students actually interpret hedged AI responses, the empirical question on which the argument against partial refusal ultimately rests.

The framework's transferability is also untested. Domains combining volatile regulation with severe consequences, such as tax guidance, benefits eligibility, and occupational licensing, share the structural features that motivate AADD, and whether the classification generalises is an open question.

10. Conclusion

The advisory journey of an internationally mobile student begins with aspiration and succeeds only in belonging. Artificial intelligence can assume a meaningful share of the work in between, but only if deployment is governed by the right variable. This chapter has argued that the variable is not how difficult a question is, but how quickly its answer changes and how much damage a wrong answer does. The AADD framework operationalises that distinction across the five stages of the mobility lifecycle, and its most important provision is a category of question that AI systems should decline to answer at all. Restraint of this kind is not a limitation on the technology's usefulness. In a domain where a fluent error can end a student's presence in a country, it is the precondition of it.

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